

Pulmonary Physiology

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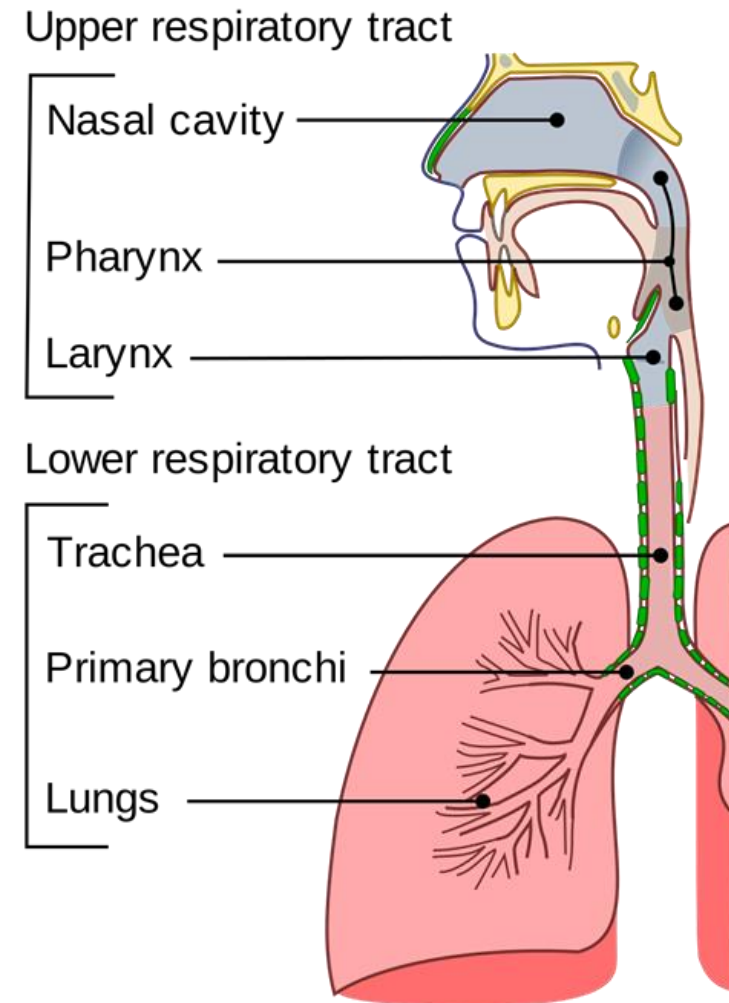
Preview

- ❑ Respiratory function of nose, trachea and bronchi, cough reflex and sneezing reflex & bad effects of smoking
- ❑ Gas exchange at the lungs and at the tissues
- ❑ Transport of oxygen and carbon dioxide in the blood
- ❑ Regulation of respiration
- ❑ Hypoxia and cyanosis

Physiologic Anatomy of Respiratory System

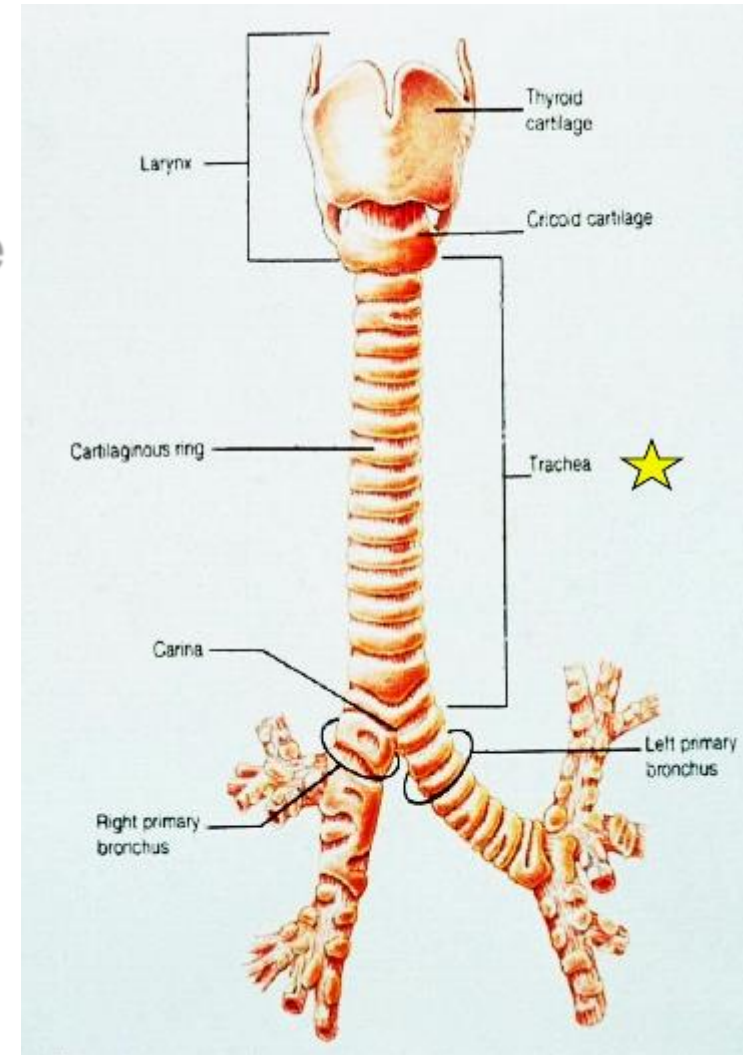
It is composed of:

- **Nasal cavity**
- **Pharynx**
- **Larynx**
- **Trachea**
- **Bronchi and bronchioles**
- **Alveoli**



Physiologic Anatomy of Respiratory System

- ✓ All the respiratory passages are kept open by incomplete cartilaginous rings except the terminal bronchioles that lack those rings
- ✓ They are also containing smooth muscles that are affected by hormones





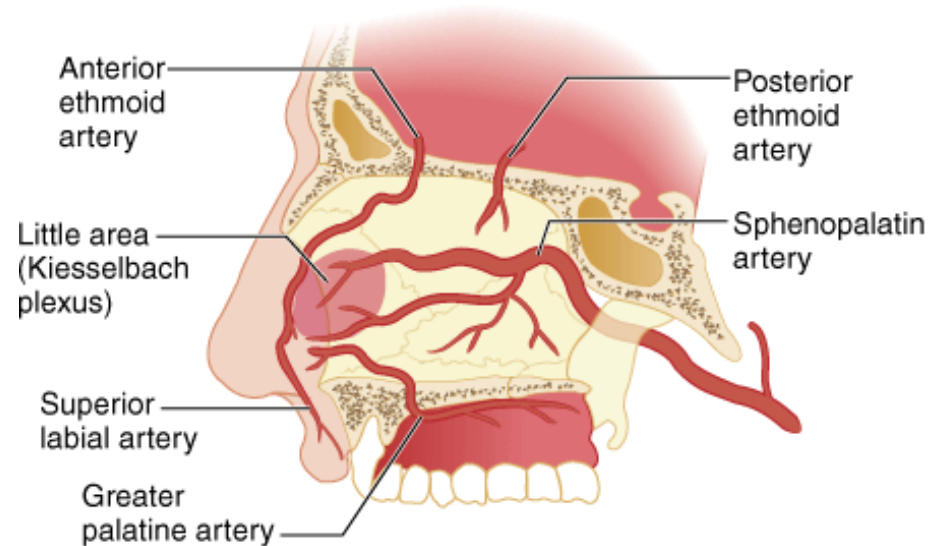
Function of the nose

It acts as resonating chamber for speech
It detects odors in the airstream

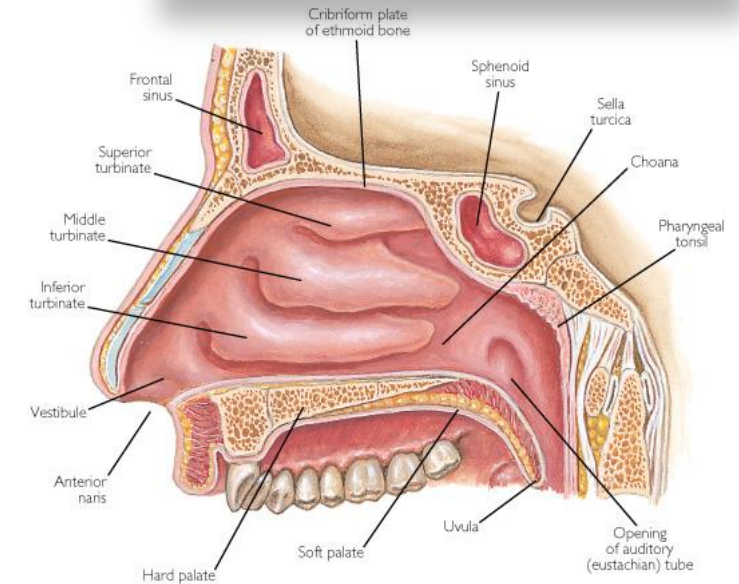


It filters the inspired air

It is an airway for respiration



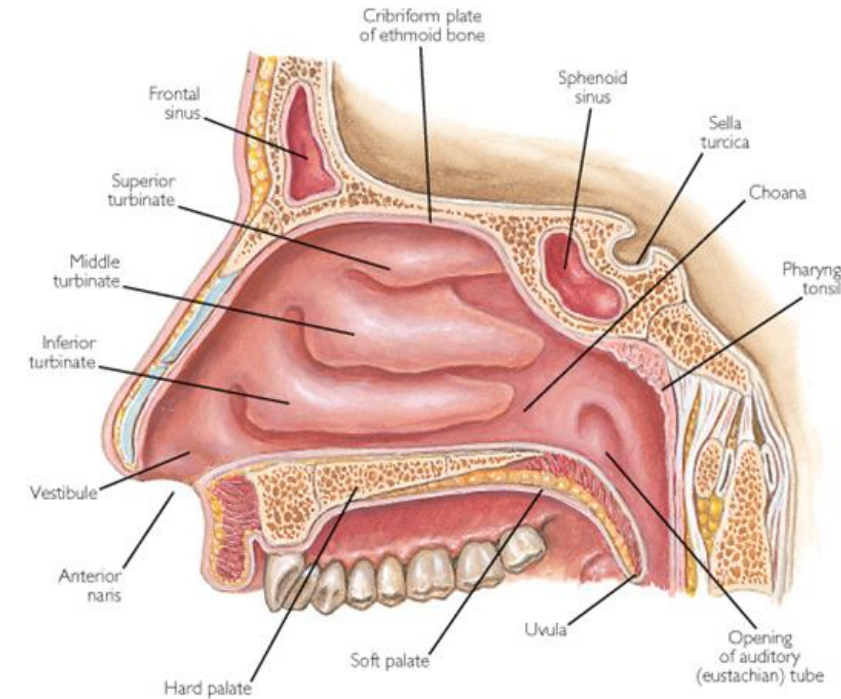
It moistens and warms the inspired air



It cleans the inspired air by nasal turbulence mechanism

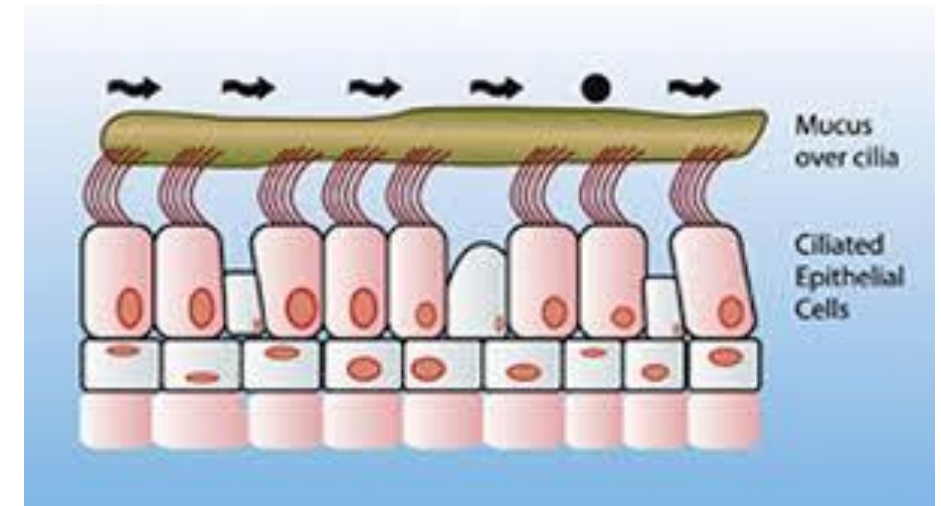
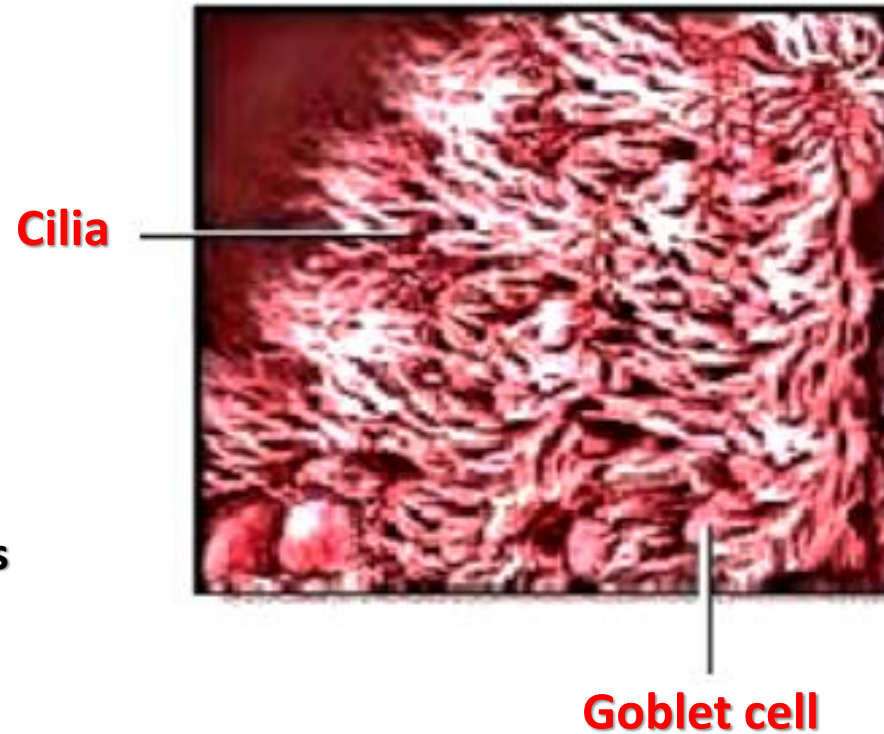
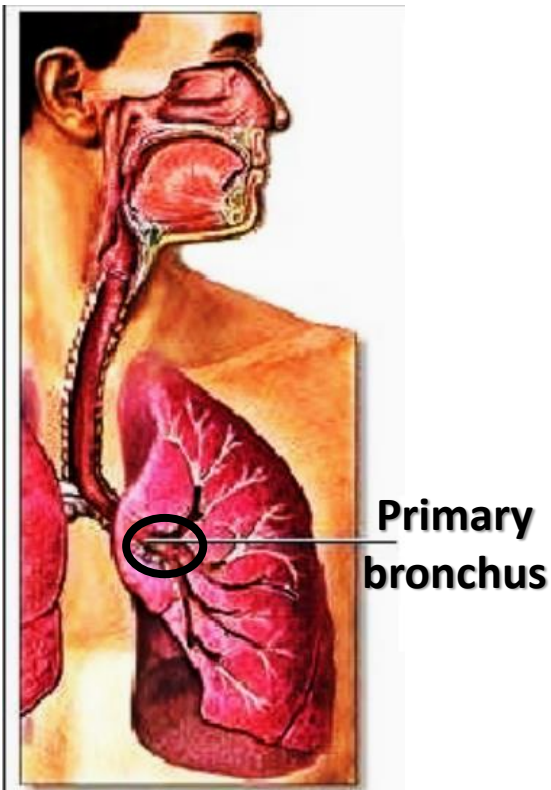
Nasal turbulence is so effective that no particle larger than 6 μm in diameter is allowed to enter the lungs through the nose

- ❑ Size between 1 – 5 μm settle in the smaller bronchioles due to gravitational precipitation
 - For example, terminal bronchiolar disease is common among coal miners because of settled dust particles
- ❑ Size 1 μm adhere to the alveolar fluid covering the wall of the alveolus
- ❑ Particles less than 0.5 μm remain suspended and expelled with expiration



Cilia are tiny, hair like structures on epithelial cells linings of many parts of respiratory system that works as escalator

- ❑ Cilia will remove any foreign particles that stick to the mucous lining that is secreted by goblet cells



Unfortunately, the size of nicotine molecules is very tiny 0.3 μm

- ❑ It escapes all the previous protecting mechanisms and reach the alveoli
- ❑ It is either
 - 1/3rd adhere to alveolar wall & removed by the macrophages that will die and become unable to do their function properly
 - or
 - 2/3rd are suspended in the alveolar air to be expelled with expiration
- ❑ It causes irritation of the mucous membrane covering the air passages causing excess mucous secretion and morning cough
- ❑ At the same time it causes paralysis of the cilia



IS NICOTINE
REALLY THAT
BAD?

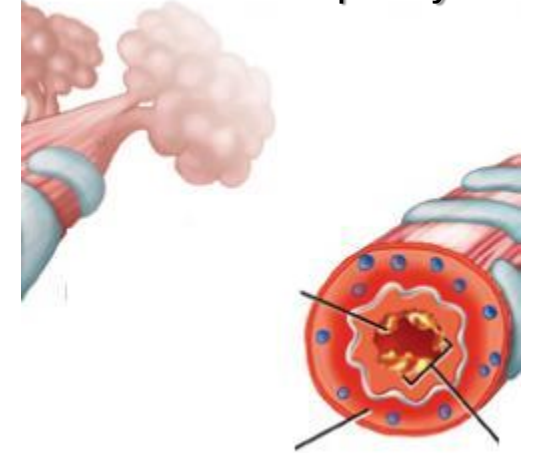


Healthy lungs

Healthy alveoli

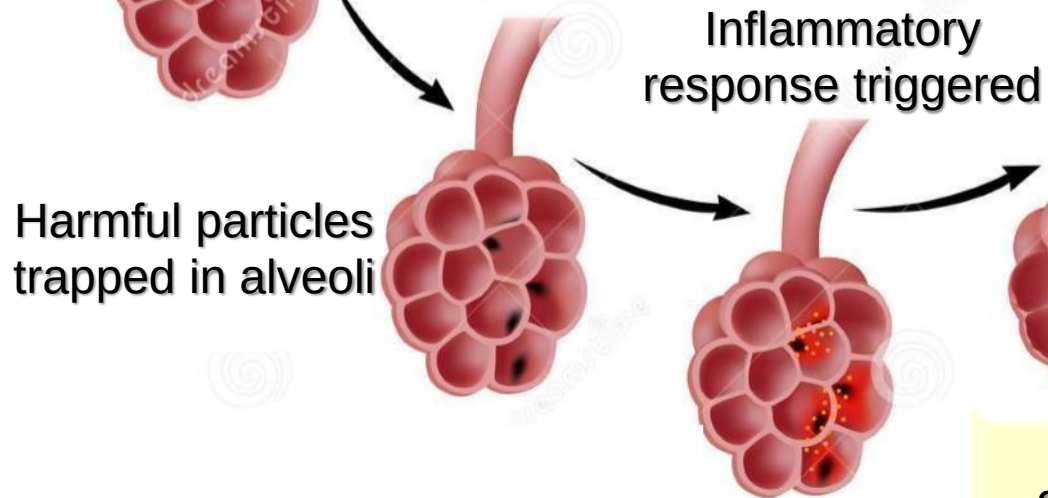
Smoking & emphysema

Smoking causes augmentation of mucous production and cilia paralysis



Inflammatory reaction partially blocks the terminal bronchioles during expiration

Air trapping in the alveoli causing rupture of the alveoli



Large air cavity lined with carbon deposits

Inflammatory chemicals dissolve alveolar septum

